KhelQuest: AI-Powered Platform for Democratizing Sports

A Mini-Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

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Acknowledgements

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **Sowmya M**, Assistant Professor and Project Supervisor, Department of Computer Science and Engineering, Vardhaman College of Engineering, for his able guidance and useful suggestions, which helped us in completing the project in time.

We are particularly thankful to **Dr. Gagandeep Arora**, the Head of the Department, Department of Computer Science and Engineering, his guidance, intense support and encouragement, which helped us to mould our project into a successful one.

We show gratitude to our honorable Principal **Dr. J.V.R. Ravindra**, for providing all facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartful thanks to **Dr. Teegala Vijender Reddy**, Chairman and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing a congenial atmosphere to complete this project successfully.

We also thank all the staff members of Computer Science and Engineering department for their valuable support and generous advice. Finally thanks to all our friends and family members for their continuous support and enthusiastic help.

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Abstract

The opportunity to get organized sports is still there. unequal because of the geographical, resource and visibility restrictions, especially among upcoming sports stars and local communities. This essay describes KhelQuest, a digital platform that is AI-powered form that would democratize sports through the association of players, teams, managers, and organizations in an integrated ecosystem. The platform uses artificial intelligence to facilitate talent, discovery, custom training insights, event discovery and performance tracking, which decreases conventional obstacles related to the price, place and availability of professional networks. Integrating evidence-based suggestions and community. participation, KhelQuest is favorable to talent detection and enables fairness in talent detection. evidence-based decision-making in favor of both athletes and stakeholders. The suggested system promises inclusiveness, scalability, and transparency, which is meant to alter the way sports opportunities are accessed and managed. The conceptual idea is described in this work. structure, fundamental features and possible impact on society. KhelQuest, emphasizes that it helps in equal opportunity and sustainable development of sports ecosystem.

Keywords: Artificial Intelligence, Sports Talent Identification, Computer Vision, Smartphone-Based Performance Analysis, Machine Learning, Sports Analytics

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Abbreviations

Abbreviation	Description
VCE	Vardhaman College of Engineering
AIML	Artificial Intelligence and Machine Learning

CHAPTER 1

Introduction

1.1 Introduction

The KhelQuest project is an AI-powered platform designed to democratize access to sports opportunities by enabling efficient and accessible talent identification. The system utilizes smartphone-based video analysis, computer vision, and machine learning techniques to evaluate athletic performance without requiring expensive infrastructure or specialized equipment.

Sports play a crucial role in physical development, social interaction, and national progress. However, access to structured training and talent identification remains limited due to geographical, economic, and infrastructural barriers. KhelQuest addresses these challenges by providing an inclusive and scalable digital solution that connects athletes, coaches, and organizations within a unified ecosystem.

The primary aim of this project is to develop a system that provides accurate, unbiased, and real-time performance evaluation. The scope of the project includes mobile-based data acquisition, AI-driven analysis, and cloud-based performance tracking. This report presents the design, implementation, and evaluation of the proposed system in a structured manner.

1.2 Background and Motivation

The field of sports analytics has evolved significantly with advancements in artificial intelligence, computer vision, and mobile computing. Traditional methods of talent identification rely heavily on manual observation, expensive equipment, and centralized training facilities, which limits participation, especially in rural and underprivileged regions.

Recent developments in AI technologies, such as pose estimation and activity recognition, enable detailed analysis of human movement using video data.

These techniques allow performance evaluation without the need for wearable sensors or laboratory setups. However, most existing systems are designed for professional environments and lack scalability for grassroots applications.

The motivation behind this project is to bridge this gap by developing a mobile-first, AI-based platform that provides accessible and fair sports analytics. KhelQuest aims to eliminate barriers related to cost, location, and availability of resources, thereby promoting equal opportunities for athletes across diverse backgrounds.

1.3 Objectives of the Project Work

The main objective of this project is to design and develop an AI-based system for sports performance evaluation and talent identification. The specific objectives of the project are as follows:

1. To develop a smartphone-based system for capturing and analyzing athletic performance using video data.
2. To implement computer vision and machine learning techniques for accurate movement analysis and performance evaluation.
3. To design a fair and unbiased scoring mechanism for talent identification across different user groups.
4. To develop lightweight and efficient AI models suitable for deployment on resource-constrained devices.
5. To provide real-time feedback and performance insights to athletes.
6. To integrate cloud-based storage for tracking performance history and enabling data-driven decision-making.

1.4 Organization of the Report

This report is organized into several chapters to present the project in a structured manner:

- **Chapter 2**: Literature Review – Discusses existing research and technologies related to AI in sports analytics and highlights research gaps.
- **Chapter 3**: System Design – Describes the architecture, components, and overall design of the proposed KhelQuest platform.
- **Chapter 4**: Methodology and Implementation – Explains the working process, algorithms, and implementation details of the system.
- **Chapter 5**: Results and Discussion – Presents the experimental results and provides analysis and interpretation of the findings.
- **Chapter 6**: Conclusion and Future Scope – Summarizes the project outcomes and discusses potential future enhancements.

CHAPTER 2

Literature Survey

2.1 Introduction

In this chapter, an overview of the existing literature related to the topic of artificial intelligence in sports analytics and talent identification is discussed. The objective of conducting this literature survey is to understand the current state of technology and methodologies used in the field. At the same time, it also helps to identify the limitations of the technology and establish the need for the proposed KhelQuest platform.

In conducting the literature survey, relevant research papers on machine learning, computer vision, pose estimation, and cloud-based sports analytics platforms were reviewed. This chapter is divided into two sections. The first section is on the review of prior research, and the second section is on the identification of research gaps.

2.2 Review of Prior Research

It is found that the application of artificial intelligence in the domain of sports analytics is significant, as it helps in the evaluation and analysis of the performance of athletes and the identification of talents. There are various studies that have shown the application of machine learning in the analysis and evaluation of the performance of athletes, making the process more consistent and less subjective.

Computer vision is an important application of artificial intelligence, and the application of the concept of ”pose estimation” is significant in the analysis and evaluation of the movement of humans. The application of the concept helps in the estimation and analysis of the skeletal joints, and the application of the concept is found to be accurate, as shown in the studies, even when the application is done using consumer-grade devices.

The application of lightweight machine learning and edge computing is significant, as the application helps in the deployment and execution of the concept of artificial intelligence on mobile devices, making the application and analysis of the concept in the domain of sports analytics feasible. The application of the concept is significant, as the application is feasible and can be done on mobile devices.

The application and analysis of the concept of cloud-based sports management are significant, as the application helps in the evaluation and analysis of the performance of athletes, and the application is feasible and can be done using the concept. However, the application and analysis of the concept are feasible only in the professional domain and are not accessible to the grassroots domain.

In addition, the application and analysis of the concept have shown the importance and need to consider the ethical implications and issues related to the application and analysis of the concept, as shown in the studies, and the application and analysis of the concept have shown the importance and need to consider the security issues and aspects related to the application and analysis of the concept, as shown in the studies.

2.3 Identified Research Gaps

Despite all these advances in sports analytics using AI, there are some limitations associated with existing systems. Most of the existing sports analytics systems have been developed for professional sportsmen and require either a controlled environment or expensive equipment, making it difficult for sportsmen in rural areas to be part of such systems.

Most sports analytics systems have been developed for analyzing sports performance only, without considering all stakeholders like sportsmen, coaches, and administrators under one umbrella. These limitations have been identified for effective sports talent development and management.

Another major limitation of existing sports analytics systems is that they have not been designed for low-resource conditions and have failed to address issues like limited connectivity, devices, and environmental variations.

Moreover, issues like fairness, bias, and ethical issues associated with sports analytics using AI have not been addressed properly in existing systems. In addition, reliable and efficient systems for preventing manipulation of sports performance data have not been implemented effectively.

All these limitations have been identified for developing an effective sports analytics system based on AI, mobile technology, and cloud computing for sports talent development.

2.4 Summary

This chapter presented the existing literature in the domain of AI-based sports analytics, with emphasis on important technologies such as ML, computer vision, and cloud computing. Although considerable advancements have been achieved, the current state of affairs is lacking in terms of accessibility, scalability, and integration.

The gaps identified in the existing literature have created the need to develop the proposed KhelQuest system, which is mobile-based and uses AI to ensure fair and efficient talent identification in the world of sports. The following chapters discuss the proposed system's design, methodology, and implementation.

CHAPTER 3

Prior Research

3.1 Introduction

In recent years, the use of Artificial Intelligence (AI) and Machine Learning (ML) in sports has gained significant attention. Traditional methods of sports talent identification rely heavily on manual observation, expensive equipment, and expert judgment. These approaches are often limited to urban areas and professional training centers, making them inaccessible to many talented athletes, especially in rural regions.

With advancements in computer vision and mobile technology, researchers have explored AI-based solutions for analyzing human movements, evaluating athletic performance, and reducing bias in decision-making. This chapter reviews existing research and technologies related to sports analytics, pose estimation, and mobile-based performance assessment. It also highlights the limitations of current systems and the need for an inclusive, scalable solution like the proposed platform.

3.2 Existing Work

Several studies have focused on using computer vision and AI techniques to analyze sports performance. Vision-based motion analysis systems use video input to track body movements and evaluate performance without requiring physical sensors. Technologies such as pose estimation models (e.g., keypoint detection) enable accurate measurement of exercises like squats, push-ups, and jumps.

Research shows that these systems can provide reliable results comparable to traditional biomechanical tools. Additionally, machine learning algorithms have been used to automate performance evaluation, reducing human bias and improving consistency.

Table 3.1: Comparative Analysis of Existing Works and Proposed System

Ref.	**Methodology**	**Technology**	**Deployment**	**Limitation**	**KhelQuest Advantage**
[1]	Survey-based review of AI in sports	General ML and analytics models	Conceptual research studies	No real-world system implementation	Provides practical mobile-based system
[2]	Video-based ML performance evaluation	Computer vision + ML classification	Controlled indoor setups	Limited scalability discussion	Works with smartphone videos in real-world conditions
[3]	Deep learning for talent detection	CNN/DNN trained on labeled datasets	Structured dataset environments	High dataset dependency	Operates on real-world recordings without strict dataset constraints
[4]	Lightweight model optimization for edge devices	TinyML and quantized networks	Low-resource embedded devices	Performance trade-offs in scaling	Combines lightweight AI with cloud support for better efficiency
[5]	Cloud-based sports analytics dashboards	Web systems + database analytics	Professional sports institutions	Not designed for grassroots usage	Targets rural and decentralized athletes using mobile platform
[6]	Secure data management framework	Encryption and cloud authentication	Mobile health applications	Not sports-specific	Integrates secure data handling into sports performance system
[7]	Fairness and bias analysis in AI systems	Bias detection and transparency models	Theoretical research focus	Limited practical implementation	Implements fairness and transparency in athlete evaluation
Proposed	End-to-end AI sports assessment pipeline	Pose estimation + ML + Cloud + Mobile integration	Grassroots and decentralized environments	Performance may vary based on video quality and conditions	Provides an integrated, scalable, and inclusive ecosystem for talent identification

3.3 Mobile-Based Solutions and Limitations

Recent developments have introduced mobile-based applications that use lightweight AI models for activity recognition and fitness tracking. Frameworks such as MediaPipe and TensorFlow Lite allow real-time pose detection on smartphones, making AI more accessible to the general population.

These solutions demonstrate that smartphones can be effectively used for sports performance analysis at a lower cost. However, many existing applications are limited to basic fitness tracking and do not provide comprehensive features such as athlete networking, verified performance records, ranking systems, or connections with coaches and sports authorities.

Furthermore, issues such as lack of offline functionality, limited personalization, and absence of structured training guidance still exist. There is also a gap in integrating AI analysis with real-world opportunities like talent selection and mentoring.

3.4 Summary

The literature review indicates that while significant progress has been made in applying AI and computer vision to sports analytics, existing systems are often expensive, complex, and not widely accessible. Mobile-based solutions have improved accessibility but still lack integration, scalability, and real-world impact.

These gaps highlight the need for a unified platform that combines AI-based performance analysis, personalized training, athlete networking, and opportunities for talent recognition. The proposed system aims to address these limitations by providing an inclusive, cost-effective, and scalable solution for democratizing sports talent identification.

CHAPTER 4

Proposed Work

4.1 Introduction

The proposed system, AI–Powered Platform for Democratizing Sports, aims to transform traditional sports talent identification by leveraging Artificial Intelligence, mobile technology, and cloud computing. Unlike conventional methods that rely on expensive equipment and manual evaluation, this system enables athletes to perform standardized fitness tests using smartphones.

The platform is designed to be inclusive, scalable, and cost-effective, allowing athletes from both urban and rural areas to participate. It not only evaluates performance using AI-based analysis but also supports improvement through personalized training plans, feedback mechanisms, and connections with coaches and sports authorities. The system integrates multiple modules such as athlete profiling, AI-based performance assessment, ranking systems, and real-world opportunity access into a single unified platform.

4.2 Architecture

The system follows a modular architecture consisting of multiple interconnected components:

4.2.1 Athlete Mobile Application

User registration and profile creation, Video/image capture for performance tests, Access to dashboard, training plans, and feedback

4.2.2 AI Processing Module

Pose detection using computer vision, Repetition counting (push-ups, squats, jumps), Cheat detection and anomaly analysis, Performance scoring (speed,

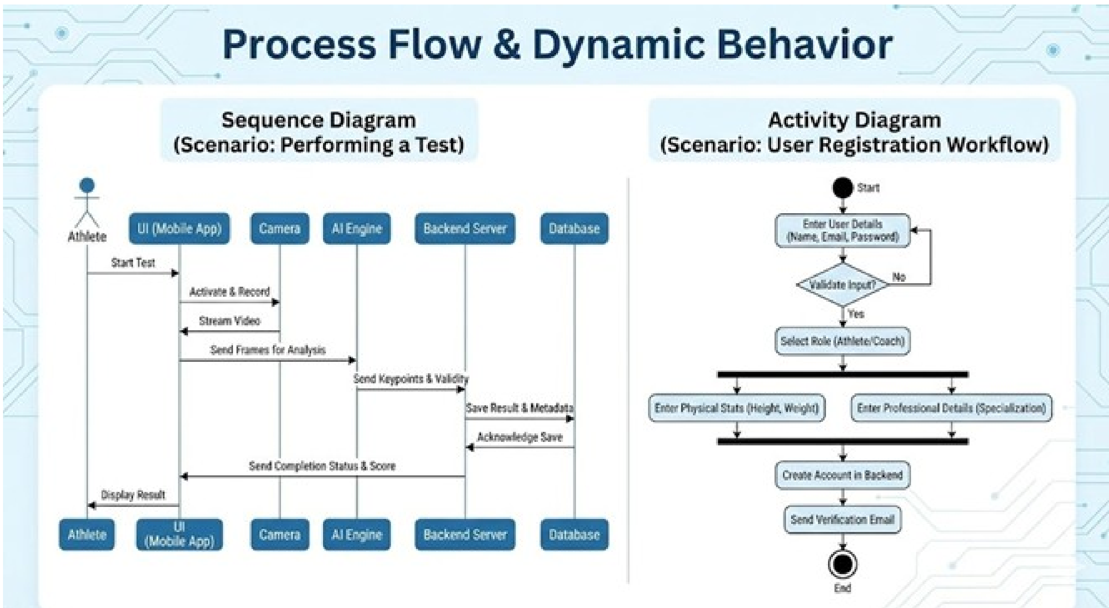

Figure 4.1: System Design

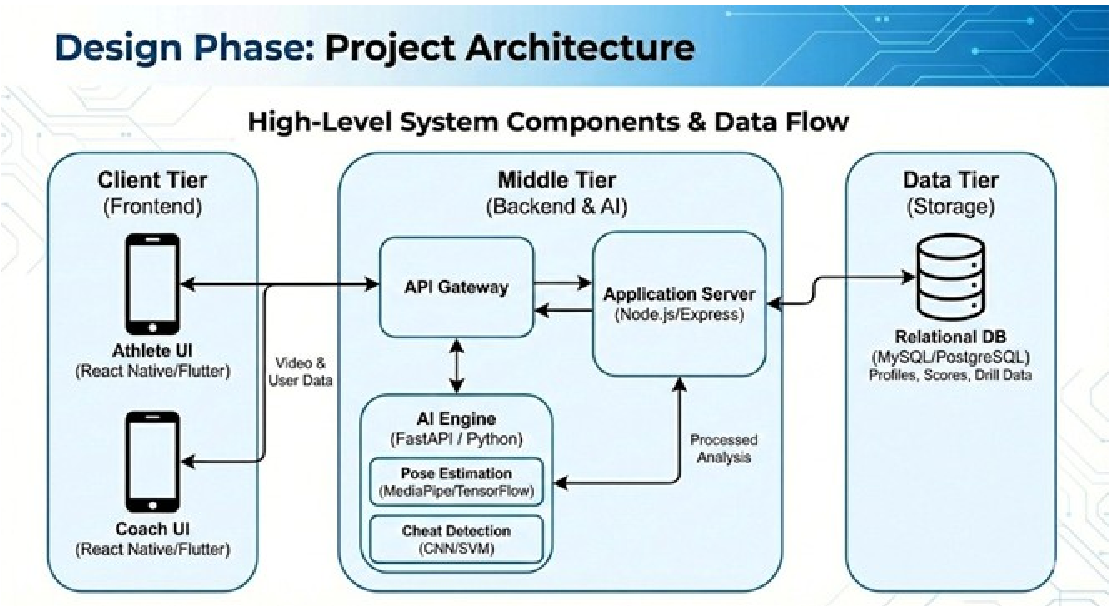

Figure 4.2: Architecture

endurance, technique, consistency)

4.2.3 Backend Server

API services for communication, Authentication and user management, Data processing and integration,

4.2.4 Database

Stores athlete profiles, performance data, rankings Maintains secure and verified records

4.2.5 Analytics and Dashboard

Displays performance metrics, Tracks goals and milestones, Shows rankings (district, state, global)

4.2.6 Additional Modules

AthleteConnect (peer interaction) Playground Finder (booking and availability), Coach Module (real coach + AI coach), Sports Authority Integration

4.3 Methodology

The system follows a step-by-step methodology for performance evaluation and improvement:

4.3.1 User Registration and Profiling

Athletes begin by creating personal accounts within the platform. During registration, they provide essential personal details (such as age, gender, and contact information) along with sport-specific information (including discipline, training level, and competitive history).

4.3.2 Test Selection and Data Capture

Once registered, athletes select their sport from a predefined list. Based on the chosen discipline, the system generates a set of fitness and performance tests relevant to that sport. Athletes then perform these tests under guided instructions provided by the application.

4.3.3 AI-Based Analysis

The captured video is processed using pose estimation models. The system extracts body keypoints, evaluates posture, counts repetitions, and calculates performance metrics.

4.3.4 Performance Evaluation

The system generates scores based on parameters such as speed, endurance, technique, and consistency.

4.3.5 Feedback and Suggestions

AI provides corrective feedback and identifies areas for improvement, including injury risk detection.

4.3.6 Training Plan Generation

Personalized training plans are created based on the athlete's performance level and goals.

4.3.7 Data Storage and Ranking

Results are stored securely in the database and used to generate rankings at different levels.

4.3.8 Connection and Opportunities

Athletes can connect with peers, coaches, and sports authorities for mentorship and selection opportunities.

4.4 Implementation

The implementation of the proposed system is carried out using modern technologies:

4.4.1 Frontend

1)Mobile/Web interface using React, HTML, CSS, and JavaScript

2)User-friendly interface for easy navigation

4.4.2 AI/ML Module

1)MediaPipe for pose detection

2)TensorFlow Lite for on-device processing

3)Machine learning models for scoring and cheat detection

4.4.3 Backend

1)Node.js and Express for API development

2)FastAPI for AI service integration

4.4.4 Database

1)MySQL for structured data storage

2)Firebase Firestore for real-time updates

4.4.5 Deployment

1)Cloud-based deployment for scalability

2)CI/CD using GitHub Actions

4.4.6 Key Implementation Features

1)Offline video capture with delayed upload

2)Lightweight models for low-end devices

3)Secure authentication and data encryption

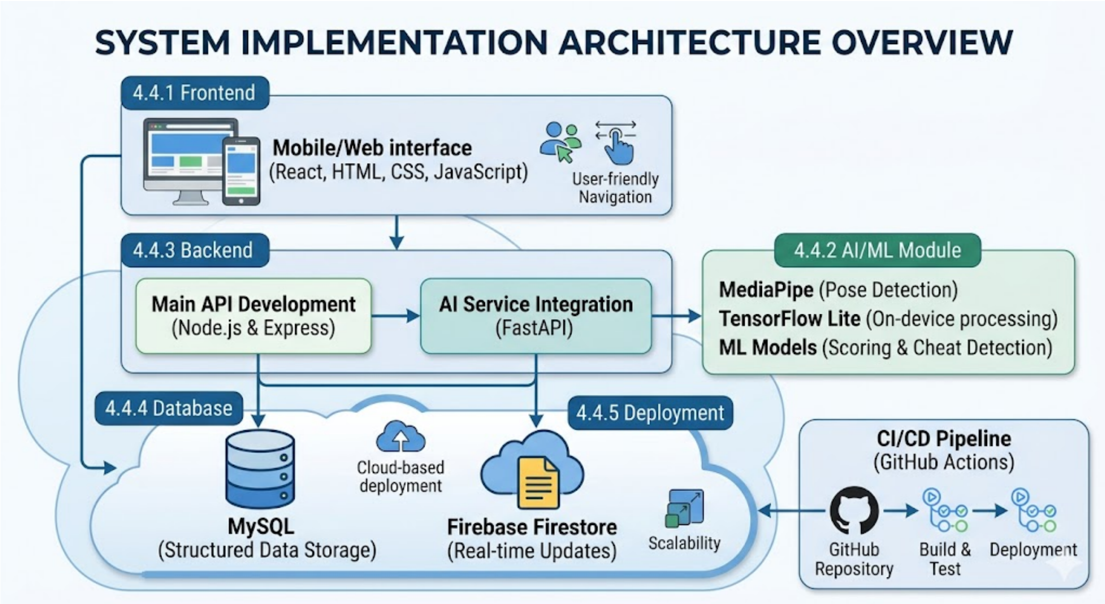

Figure 4.3: System Implementation

4.5 Summary

The proposed system provides a comprehensive solution for sports talent identification by combining AI, mobile technology, and cloud computing. It addresses the limitations of existing systems by offering accessibility, affordability, and transparency.

Through features such as AI-based performance analysis, personalized training plans, athlete networking, and integration with sports authorities, the platform creates a complete ecosystem for athlete development. The system is designed to be scalable and adaptable, making it suitable for nationwide implementation and contributing to the democratization of sports talent identification.

CHAPTER 5

Results and Discussion

5.1 Introduction

In this chapter, the results obtained from the implementation of KhelQuest are presented, and a detailed discussion on the results obtained from its implementation is provided. The results obtained from the implementation of KhelQuest prove the effectiveness of the proposed AI-based system in assessing sports performance through video analysis using a smartphone. This chapter has been divided into different sections, namely, an overview of the results, analysis and discussion, evaluation based on quality factors, and a conclusion.

5.2 Overview of Results

The efficacy of the KhelQuest platform was tested with various fitness measures such as push-ups and vertical jumps carried out by athletes at different age groups and locations. The system was able to process video inputs through mobile devices and apply AI-based models.

The platform produced performance scores and rankings, as well as real-time feedback without the use of additional sensors or manual evaluation methods. It also provided multi-level rankings such as age group and region-wise comparisons to help the athletes compare their performances with others in the same group. From the results obtained, it was observed that the system provided consistent response times and could store the historical performances of the athletes.

5.3 Analysis and Interpretation

The results show that the application of artificial intelligence, computer vision, and mobile technologies is useful in the accurate and unbiased evaluation

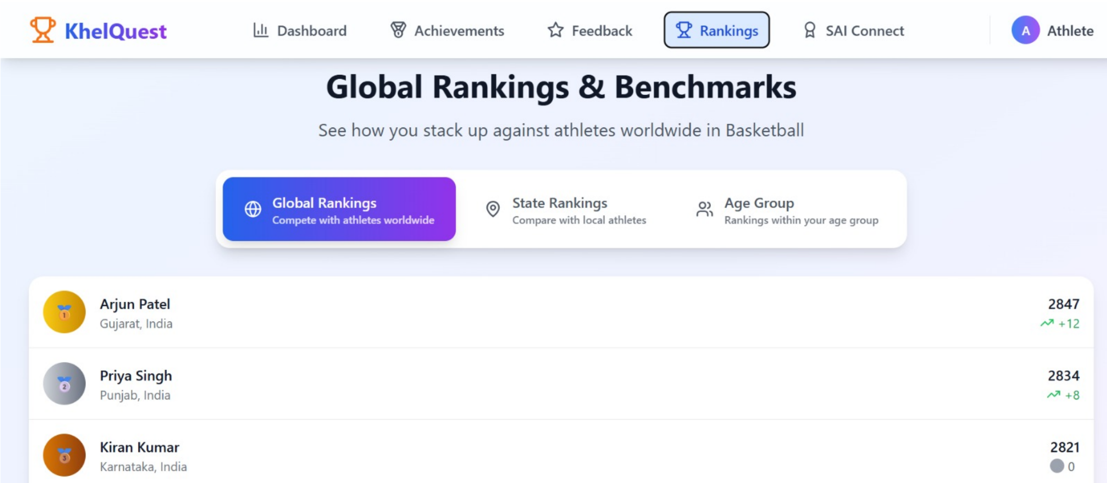

Figure 5.1: Dashboard

of sports performance. The system is efficient in the analysis of the movement patterns and the application of standardized scores.

The differences in the performance scores are attributed to the precision and quality of the movement patterns. The application of the ranking system promotes the spirit of competition and motivation among the athletes. The system is an improvement over the traditional methods, as it is cost-effective and applicable in the grassroots.

The application of cloud-based analytics is also useful in the monitoring and evaluation of the progress of the athletes. The results have achieved the objectives of the project, which are focused on the provision of accessible, unbiased, and efficient evaluation methods.

5.4 Evaluation of Quality Factors

- **Environment:** The system reduces the need for physical infrastructure and dependence on resource-intensive training facilities. By facilitating digital assessment, the system helps to promote an environmentally efficient approach.

- **Sustainability:** KhelQuest is developed as an scalable and sustainable solution. The mobile-based solution and cloud support ensure the sustainability of the solution in regions, especially the rural areas.

- **Safety:** The system facilitates the safe use of the equipment through the avoidance of complex equipment. The users engage in physical exercises in familiar environments, which are safer.

- **Ethics:** The system ensures fairness and transparency through the use of automated scoring systems. Data privacy is ensured, which facilitates the ethical use of technology.

- **Cost:** The solution is highly cost-effective as it eliminates the need to establish costly infrastructures, wearable devices, and training systems.

- **Type:** The results are consistent with the objectives of developing an AI-driven sports analytics system. The system is applicable in real-world scenarios and can contribute to the development of sports at the grassroots level.

- **Standards:** The system uses standardized evaluation metrics and ensures consistency in the assessment of the system's performance, which is consistent with the practices of sports analytics systems.

5.5 Summary

As shown in this chapter, the KhelQuest platform is a viable tool for the accurate evaluation of sports performance with the use of AI and mobile technology. This validates the use of the system in real-world settings and its potential to change the face of talent discovery and development of sporting talent in an inclusive manner.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

The KhelQuest platform offers a new and innovative approach to democratizing sports talent identification through the use of AI and mobile technologies. The platform leverages video analysis, computer vision, and machine learning technologies on smartphones, allowing for accessible performance evaluation for athletes from different socio-economic and geographical backgrounds, thereby eliminating the need for costly infrastructures.

The proposed solution effectively addresses various problems associated with traditional sports infrastructures, such as subjective evaluation, inaccessibility, and centralization of sports talent identification. The use of performance scoring and evaluation metrics allows for a fair and transparent approach to sports talent identification through AI and machine learning technologies. The use of lightweight AI models also allows for efficient and effective implementation, especially in rural areas and developing countries.

Additionally, the use of cloud-based analytics allows for long-term performance evaluation and analysis, providing a powerful platform for coaches and sports administrators to make data-driven decisions. The platform has tremendous potential for improving athlete engagement, training efficiency, and developing a robust sports ecosystem.

KhelQuest fills the gap between sports analytics and accessibility, providing a digital platform for democratizing sports and bridging the gap between accessibility and innovation, providing a powerful platform for equal opportunities and growth in the sports ecosystem.

6.2 Future Scope of Work

The KhelQuest platform offers significant potential for future enhancements and expansion. Several directions can be explored to further improve its capabilities and impact on the sports ecosystem.

1. **Multi-Sport Expansion:** The current system can be extended to support a wider range of sports by incorporating sport-specific movement models and evaluation metrics. This would enable broader adoption across different athletic disciplines.

2. **Advanced Predictive Analytics:** Future improvements can include predictive modeling techniques to analyze historical performance data and forecast athlete development. This can assist in early talent identification and long-term training planning.

3. **Wearable and Multimodal Integration:** Integrating wearable sensors such as accelerometers and heart-rate monitors can enhance performance analysis. Combining visual data with physiological data can provide more accurate and comprehensive insights.

4. **Scalability and Regional Deployment:** Enhancing cloud infrastructure and adding multilingual support can enable large-scale deployment across different regions. This would make the platform more accessible to diverse populations.

5. **Explainable and Ethical AI:** Incorporating explainable AI techniques can improve transparency in automated scoring systems. Ensuring fairness, privacy, and bias reduction will be critical for responsible AI deployment.

6. **Enhanced Anti-Cheating Mechanisms:** Future systems can include advanced anomaly detection and video verification techniques to prevent manipulation or fraudulent submissions, ensuring system integrity.

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